CUSTOMER SHOPPING BEHAVIOR ANALYSIS

1. Project Overview

This research uses transactional data from 3,900 purchases made in a variety of product categories to examine consumer purchasing behavior. Finding insights on consumer segments, product preferences, spending trends, and subscription behavior is the aim in order to inform strategic business decisions.

2. Dataset Summary

1. 3,900 rows.
2. 18 columns.

Important Features:

1. Demographics of the client (Age, Gender, Location, Subscription Status).
2. Purchase information (Item Purchased, Category, Purchase Amount, Season, Size, Color) .
3. Purchase habits (discount applied, promo code used, prior purchases, frequency of purchases, review rating, shipping type).
4. 37 data in the Review Rating column are missing.

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

1. **Data Loading:** Pandas was used to import the dataset.
2. **First Investigation:** used describe() for summary statistics and df.info() to verify structure.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN

Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3863.000000	3900	3900	3900	3900	3900.000000	3900	3900
NaN	2	6	2	2	NaN	6	7
NaN	No	Free Shipping	No	No	NaN	PayPal	Every 3 Months
NaN	2847	675	2223	2223	NaN	677	584
3.750065	NaN	NaN	NaN	NaN	25.351538	NaN	NaN
0.716983	NaN	NaN	NaN	NaN	14.447125	NaN	NaN
2.500000	NaN	NaN	NaN	NaN	1.000000	NaN	NaN
3.100000	NaN	NaN	NaN	NaN	13.000000	NaN	NaN
3.800000	NaN	NaN	NaN	NaN	25.000000	NaN	NaN
4.400000	NaN	NaN	NaN	NaN	38.000000	NaN	NaN
5.000000	NaN	NaN	NaN	NaN	50.000000	NaN	NaN

3. **Missing Data Handling:** Using the median rating for each product category, null values were checked and missing values were imputed in the Review Rating field.
4. **Column Standardization:** To improve readability and documentation, columns were renamed in snake case.
5. **Feature Engineering:**
    - Binning client ages resulted in the creation of the age_group column.
    - Purchase data was used to create the purchase_frequency_days column.
6. **Data Consistency Check:** Dropped promo_code_used after confirming if discount_applied and promo_code_used were redundant.
7. **Database Integration:** loaded the cleaned data after connecting a Python script to PostgreSQL and Data Frame for SQL investigation into the database.

4. Data Analysis using SQL (Business Transactions)

To address important business concerns, we conducted structured analysis in PostgreSQL.

1. **Revenue by Gender:** Total revenue from male and female clients was compared.

	gender text	revenue numeric
1	Female	75191
2	Male	157890

2. **High-Spending Discount Users:** These were consumers who took advantage of discounts but still spent more than the typical amount.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88
12	29	94

Total rows: 839 | Query complete 00:00:00.099

3. **Top 5 Products by Rating:** Products having the highest average review ratings were identified as the Top 5 Products by Rating.

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4. **Delivery Type Comparison:** The average purchase amounts for Standard and Express delivery were compared.

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

5. **Subscribers vs. Non-Subscribers:** Total revenue and average spend were compared by subscription status.

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

6. **Discount-Dependent goods:** Five goods with the largest percentage of discounted purchases were found.

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.00
3	Coat	49.00
4	Sweater	48.00
5	Pants	47.00

7. **Customer Segmentation:** Based on past purchases, clients were divided into three groups: New, Returning, and Loyal.

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

8. **Top 3 Products by Category:** These are the items in each category that are most frequently bought.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

9. **Repeat Purchasers & Subscriptions:** This study examined if consumers who have made more than five transactions are more inclined to become subscribers.

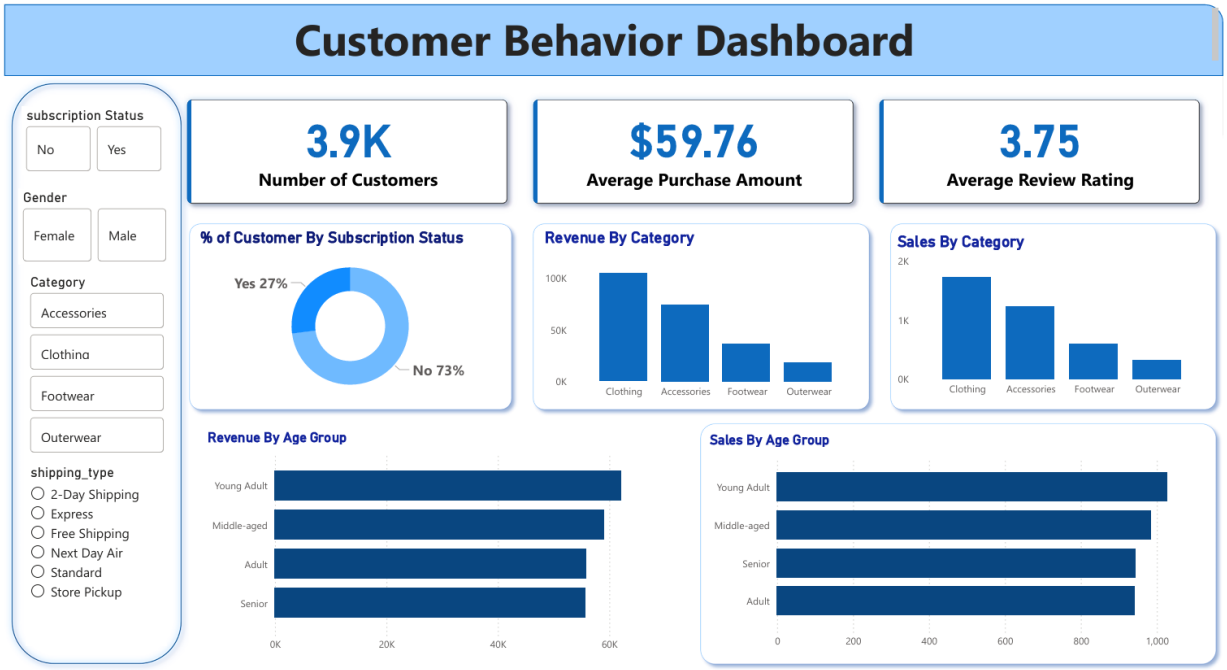
	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

10. **Revenue by Age Group:** Each age group's total revenue contribution was determined.

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

5. Dashboard in Power BI

Finally, we built an interactive dashboard in Power BI to present insights visually.



6. Business Recommendations

- **Increase Subscriptions:** Highlight special advantages for subscribers.

- **Customer Loyalty Programs:** Incentivize recurring customers to get them into the "Loyal" category.
- **Examine the discount policy:** It strike a balance between margin control and sales increases.
- **Product Positioning:** In campaigns, emphasize the best-selling and highest-rated items.
- **Targeted Marketing:** Concentrate on express-shipping customers and high-income age groups.